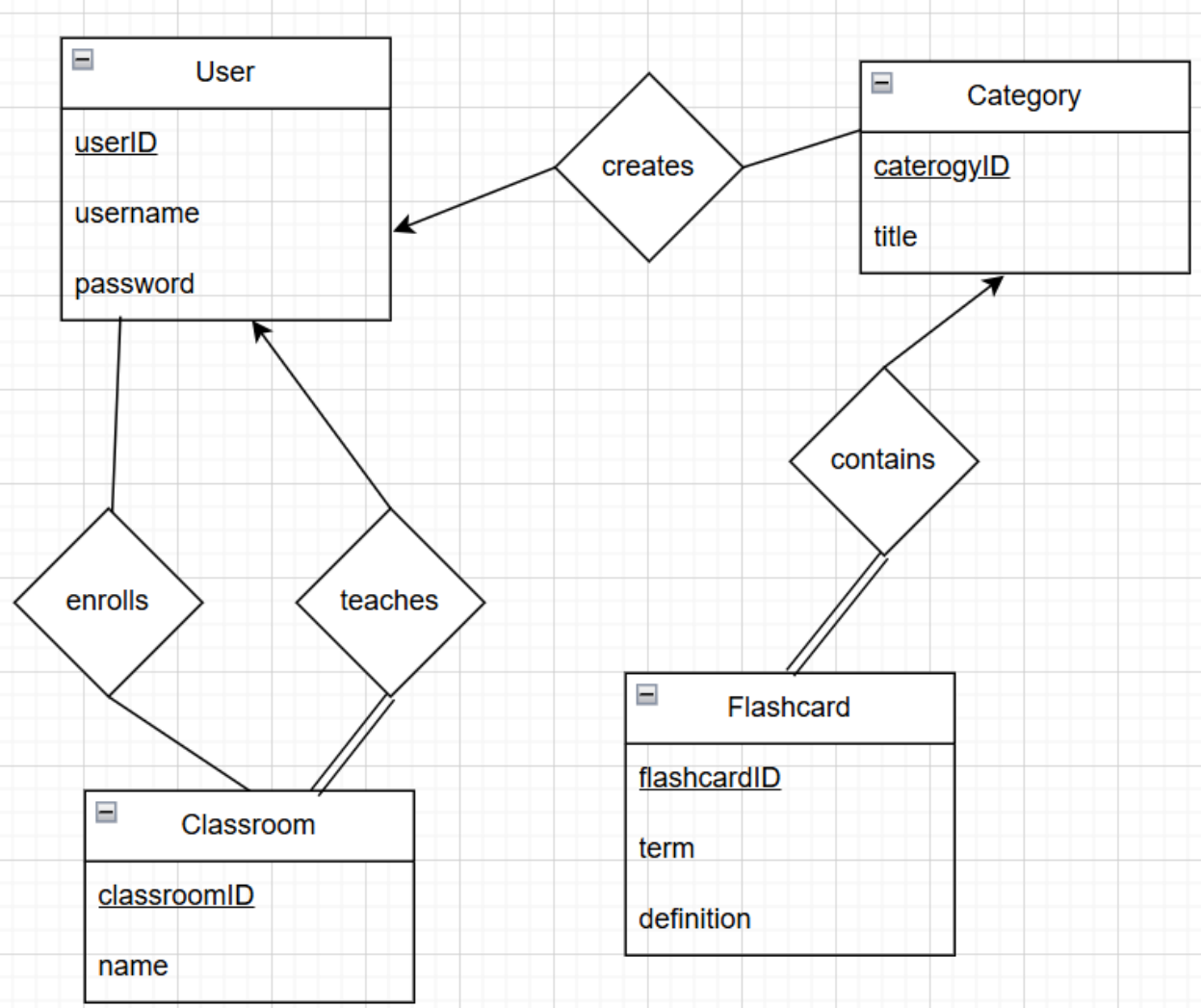
ER Model:



Resulting relations and functional dependencies after conversion:

user (

userID,

username,

password

)

pk: (userID)

unique: username

not null: username, password

Fd (user):

userID -> username, password

enrolled (

classroomID,

studentID

)

pk: (classroomID, studentID)

fk: (classroomID) references classroom(classroomID)

fk: (studentID) references user(userID)

not null: classroomID, studentID

Fd (enrolled):

classroomID, studentID -> classroomID, studentID

classroom (

classroomID,

name,

teacherID

)

pk: (classroomID)

fk: (teacherID) references user(userID)

not null: name, teacherID

Fd (classroom):

classroomID->name, teacherID

category (

categoryID,

title,

userID

)

pk: (categoryID)

fk: (userID) references user(userID)

not null: title, userID

Fd (category):

categoryID -> title, userID

flashcard (

flashcardID,

term,

definition,

categoryID

)

pk: (flashcardID)

fk: (categoryID) references category(categoryID)

not null: term, definition, categoryID

Fd (flashcard):

flashcardID -> term, definition, categoryID

Original Relations and FDs:

user(userID, username, password)

enrolled(classroomID, studentID)

classroom(classroomID, name, teacherID)

category(categoryID, title, userID)

flashcard(flashcardID, term, definition, categoryID)

userID -> username, password

classroomID, studentID -> classroomID, studentID

classroomID->name, teacherID

categoryID -> title, userID

flashcardID -> term, definition, categoryID

BCNF:

Step 1: Check all fd's to see if they are in BCNF

userID -> username, password // (userID) is the primary key

classroomID, studentID -> classroomID, studentID // (classroomID, studentID) is the primary key

classroomID->name, teacherID // (classroomID) is the primary key

categoryID -> title, userID // (categoryID) is the primary key

flashcardID -> term, definition, categoryID // (flashcardID) is the primary key

Final BCNF Relations:

user(userID, username, password)

enrolled(classroomID, studentID)

classroom(classroomID, name, teacherID)

category(categoryID, title, userID)

flashcard(flashcardID, term, definition, categoryID)